TEXORA · COURSE NOTES · LECTURE SHEET

# Theory of Dilute Solution

*The colligative properties — vapour-pressure lowering, boiling-point elevation, freezing-point depression and osmotic pressure — and what they reveal about molar mass.*

CHE-1105 · CHEMISTRY-I · TOPIC 5 · LEVEL 1 TERM 1

## 1. The colligative idea

Dissolve a non-volatile solute in a solvent and four properties change by amounts that depend — in dilute solution — only on **how many solute particles are present, not on what they are**. Such properties are called **colligative** (from "bound together" by number). The four are: lowering of vapour pressure, elevation of boiling point, depression of freezing point, and osmotic pressure. The unifying picture is entropy: solute particles dilute the solvent, lowering its escaping tendency into vapour, its readiness to freeze out, and driving it through membranes toward the solution. Because only the count matters, colligative measurements are a classical way to **determine molar masses** — weigh the solute, measure any one of the four effects, and the particle count follows.

## 2. Vapour-pressure lowering and Raoult's law

For an ideal solution, **Raoult's law** states that each component's partial vapour pressure equals its mole fraction times its pure vapour pressure. For the solvent with a non-volatile solute this reduces to:

$$p = x_{\text{solvent}} \cdot p^\circ, \quad \text{equivalently} \quad (p^\circ - p) / p^\circ = x_{\text{solute}}$$

The relative lowering of vapour pressure equals the mole fraction of the solute — the first colligative relation, and the parent of the other three (the solution boils only when its lowered vapour pressure reaches the external pressure, hence a higher boiling point; and its freezing point falls because the solid solvent separates from a diluted liquid).

## 3. Boiling-point elevation and freezing-point depression

$$\Delta T_b = K_b \cdot m \quad \text{and} \quad \Delta T_f = K_f \cdot m$$

where m is the **molality** (moles of solute per kilogram of solvent) and $K_b$, $K_f$ are the solvent's ebullioscopic and cryoscopic constants (for water: $K_b = 0.512$, $K_f = 1.86$ K kg mol$^{-1}$). Molality is used

because it is temperature-independent. Freezing-point depression is the more sensitive measurement and the classical molar-mass method.

**WORKED EXAMPLE — FREEZING POINT AND MOLAR MASS**

**(a)** 18 g of glucose (M = 180 g mol$^{-1}$, so 0.10 mol) in 500 g of water: molality = 0.10 / 0.500 = 0.20 mol kg$^{-1}$, so $\Delta T_f = 1.86 \times 0.20 = 0.372$ K — the solution freezes at **−0.372 °C**. **(b)** Run backwards: if 0.90 g of an unknown in 50 g of water depresses the freezing point by 0.186 K, then m = 0.186 / 1.86 = 0.10 mol kg$^{-1}$; moles present = 0.10 × 0.050 = 0.005 mol, so M = 0.90 / 0.005 = **180 g mol$^{-1}$**. The same arithmetic, with $K_b$, works from boiling points.

## 4. Osmosis and osmotic pressure

A **semipermeable membrane** passes solvent but not solute. Placed between pure solvent and a solution, solvent flows into the solution — **osmosis** — and the pressure that must be applied to the solution to just stop the flow is its **osmotic pressure**, π. For dilute solutions van't Hoff showed the solute behaves like a gas occupying the solution's volume:

$$\pi V = n R T, \quad \text{or} \quad \pi = c R T$$

with c the molar concentration. Osmotic pressure is by far the most sensitive colligative property (a 0.1 mol L$^{-1}$ solution at 300 K already shows π ≈ 2.46 atm), which makes it the method of choice for the huge molar masses of polymers — including the cellulose and synthetic polymers of textile science, whose chain lengths are routinely measured this way. Run in reverse — apply pressure above π to the solution — and solvent is forced out: **reverse osmosis**, the membrane technology now standard in dyehouse water recycling and effluent treatment.

## 5. Electrolytes and the van't Hoff factor

Electrolytes break the simple counting: a mole of NaCl gives nearly two moles of particles, so all four effects come out almost doubled. The correction is the **van't Hoff factor** i = (observed colligative effect) / (effect calculated for undissociated solute); every colligative formula for an electrolyte is multiplied by i. For NaCl, i ≈ 2; for $MgCl_2$, i ≈ 3; for a weak electrolyte i lies between 1 and its ion count, and from i the **degree of dissociation** follows: α = (i − 1) / (n − 1), where n is the number of ions per formula unit. The mirror-image surprise is **association**: carboxylic acids dimerise in benzene, halving the particle count, giving i ≈ 0.5 and an apparently doubled molar mass. "Abnormal" molar masses are therefore never errors — they are particle-count reports, and reading them correctly is a diagnostic of dissociation and association.

## 6. Glossary — key terms of this topic

| TERM | MEANING |
|---|---|
| Colligative property | A solution property depending on the number of solute particles, not their identity. |
| Molality (m) | Moles of solute per kilogram of solvent; temperature-independent concentration. |
| Raoult's law | Partial vapour pressure of a component = mole fraction × pure vapour pressure. |
| $K_f$ / $K_b$ | Cryoscopic / ebullioscopic constants linking freezing-point depression / boiling-point elevation to molality. |
| Osmotic pressure (π) | The pressure that stops solvent flowing through a semipermeable membrane into a solution, $\pi = cRT$. |
| van't Hoff factor (i) | Ratio of the observed colligative effect to that expected without dissociation or association. |
| Reverse osmosis | Forcing solvent out of a solution through a membrane by pressure above π; used in water purification. |

## 7. Key points and summary

- Four properties — vapour-pressure lowering, $\Delta T_b$, $\Delta T_f$, π — count solute particles; Raoult's law states the first exactly and motivates the rest.
- $\Delta T_b = K_b m$ and $\Delta T_f = K_f m$ (molality!); $\pi = cRT$ is the most sensitive and the polymer chemist's molar-mass method.
- Electrolytes multiply every effect by the van't Hoff factor i, from which the degree of dissociation follows; association ($i < 1$) betrays itself the same way.

## 8. Practice

Answer in your own words; brief model answers follow. (Water: $K_f = 1.86$, $K_b = 0.512$ K kg mol$^{-1}$; $R = 0.0821$ L atm mol$^{-1}$ K$^{-1}$.)

1. 9 g of glucose (M = 180) is dissolved in 250 g of water. Find the freezing point of the solution.
2. With the data of question 1, find the boiling point of the same solution.
3. Calculate the osmotic pressure of a 0.05 mol L$^{-1}$ sucrose solution at 27 °C.
4. The vapour pressure of pure water at a given temperature is 23.8 mmHg; a solution shows 23.0 mmHg. Find the mole fraction of the solute.

5. A 0.1 molal NaCl solution shows a freezing-point depression of 0.348 K instead of 0.186 K. Find i and the apparent degree of dissociation of NaCl.
6. Benzoic acid in benzene gives an apparent molar mass near twice its true value. Explain with the van’t Hoff factor.
7. Why is molality, not molarity, used in the ΔT formulas?
8. A dyer recycles water by reverse osmosis. State what pressure condition must be met and what passes through the membrane.

## Answer key

1. Moles = 9/180 = 0.05; molality = 0.05/0.250 = 0.20; $\Delta T_f = 1.86 \times 0.20 = 0.372$ K; freezing point = −0.372 °C.
2. $\Delta T_b = 0.512 \times 0.20 = 0.102$ K; boiling point ≈ 100.10 °C.
3. $\pi = cRT = 0.05 \times 0.0821 \times 300 = 1.23$ atm.
4. $(p^\circ - p)/p^\circ = (23.8 - 23.0)/23.8 = 0.0336$ — the solute’s mole fraction.
5. $i = 0.348/0.186 = 1.87$. $\alpha = (i - 1)/(n - 1) = 0.87/1 = 0.87$, i.e. about 87% apparent dissociation.
6. The acid dimerises (two molecules join through hydrogen bonds), halving the particle count: $i \approx 0.5$, and molar mass from $\Delta T_f$ comes out doubled.
7. Molality counts per kilogram of solvent — unaffected by temperature; molarity changes as the solution expands or contracts with temperature, while ΔT measurements span temperatures.
8. Applied pressure on the solution must exceed its osmotic pressure; solvent (water) passes through the membrane, leaving the concentrated solute (salts, dye residues) behind.

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